Supplementary Appendices

Chinook salmon spawning phenology in the Middle Fork Salmon River (CJFAS revision)

02 November 2026

Table of contents

A1 Data and study design **2**
- A1.1 Redd data and reaches . . . 2
- A1.2 Stream temperature . . . 3

A2 Why the antecedent temperature window was replaced **6**
- A2.1 Seasonal context . . . 6
- A2.2 Within a reach-year, the window re-codes the date . . . 6
- A2.3 A null model with no biology . . . 7
- A2.4 Reach-specific slopes reflect thermograph shape . . . 7

A3 Choosing a fixed-date thermal index **9**
- A3.1 Pre-spawn indices . . . 9
- A3.2 Post-spawn (incubation) indices . . . 11

A4 Other covariates and the Big Creek gradient **13**
- A4.1 Elevation . . . 13
- A4.2 Reach slope . . . 14
- A4.3 Discharge . . . 14

A5 Model structure, selection and diagnostics **15**
- A5.1 Model structure . . . 15
- A5.2 Model comparison . . . 15
- A5.3 Shape of the temperature effect . . . 15
- A5.4 Diagnostics for the final model . . . 17

A6 Final model results **19**
- A6.1 Parameter estimates . . . 19
- A6.2 Stream and year means . . . 19
- A6.3 Reach effects and position in the network . . . 19

A7 The study years in the long-term thermal record **22**
- A7.1 Were 2002–2005 typical years? . . . 22
- A7.2 Is the index a property of the reach? . . . 22
- A7.3 How large are the within-reach anomalies? . . . 22

A1 Data and study design

A1.1 Redd data and reaches

Chinook salmon redds were surveyed every 3–5 days from the onset of redd construction in late July to the end of spawning in mid-September, 2002–2005, in eight streams of the Middle Fork Salmon River. Spawn date was taken as the first date a completed redd was observed and expressed as day of year on a non-leap calendar (dates in 2004 after 28 February shifted back one day), so that the same calendar date has the same value in every year. Data from 2001 and from Knapp and Cape Horn creeks were excluded because they were not sampled consistently. The analysis dataset contains 3,016 redds in 104 reaches and 263 reach-years (Table A1.1).

Each redd location was joined to the NHDPlus Version 2 flowline network, and each flowline segment (identified by its COMID) is treated as a *reach*. Reaches are the units for which Siegel et al. (2023) predict daily stream temperature. Surveyed reaches are short: median 0.95 km (range 0.07–3.21 km). Every reach belongs to one stream, so reaches are nested within streams. 77 reaches were surveyed in two or more years and 27 in all four years; these multi-year reaches contain 2,775 of the redds and carry the within-reach (between-year) temperature contrast used in the final model (Section A5).

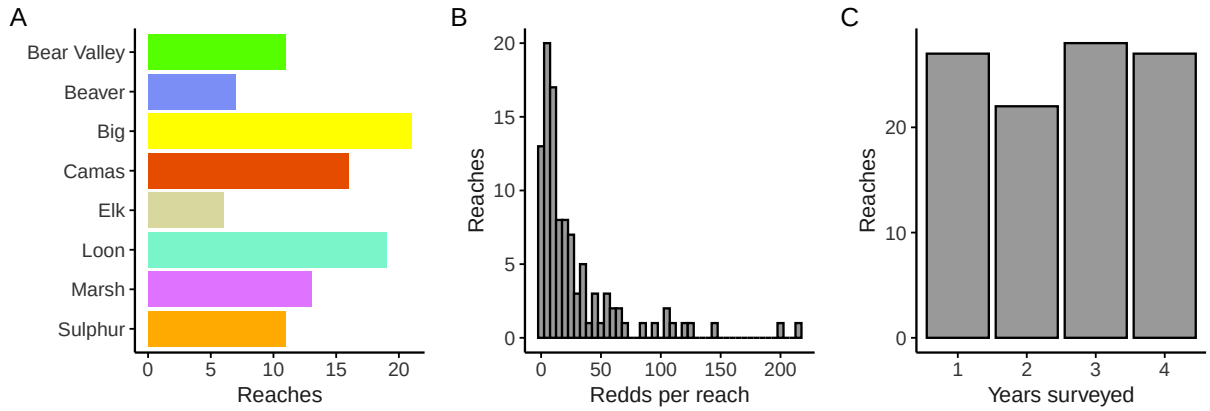


Figure A1.1: Sampling structure. (A) Number of reaches per stream. (B) Number of redds per reach. (C) Number of years each reach was surveyed.

Table A1.1: Redds by stream and year, number of reaches, and elevation range of surveyed reaches.

Stream	2002	2003	2004	2005	Redds	Reaches	Elevation (m)
Bear Valley	143	220	58	58	479	11	1940–1977
Beaver	38	141	15	9	203	7	1998–2054
Big	229	239	179	0	647	21	1166–1814
Camas	140	84	52	47	323	16	1514–1817
Elk	284	115	34	38	471	6	1970–1992
Loon	123	59	28	32	242	19	1453–1668
Marsh	122	189	33	24	368	13	1913–2015
Sulphur	108	170	5	0	283	11	1742–1779
Total	1187	1217	404	208	3016	104	

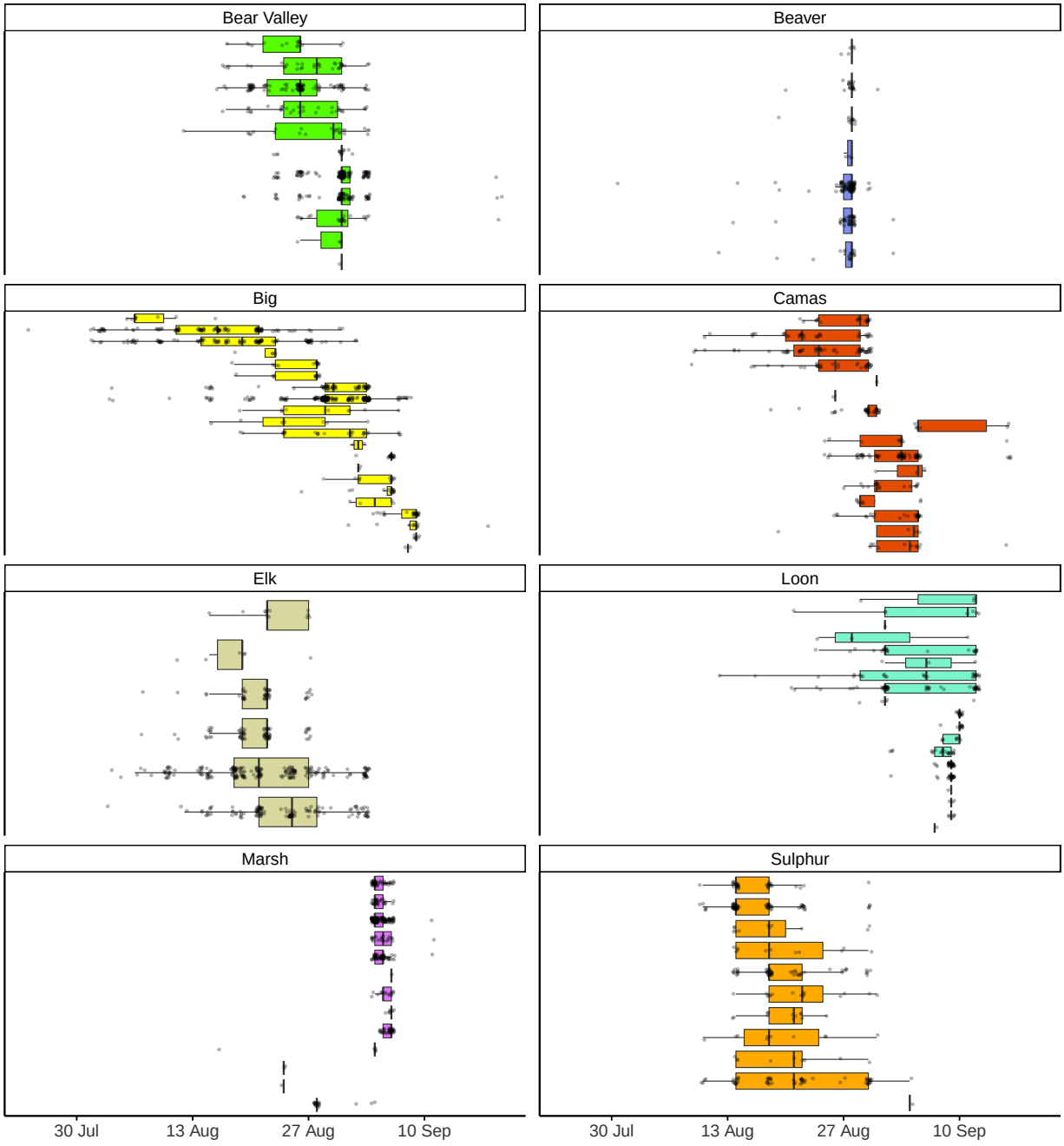


Figure A1.2: Spawn dates by reach within each stream. Boxes show the interquartile range; points are individual redds. Reaches are ordered by drainage area (upstream at top).

A1.2 Stream temperature

Daily stream temperatures were taken from the reach-scale model of Siegel et al. (2023), which agreed closely with logger data in Idaho (Figure A1.4). Figure A1.3 shows the modeled thermal regime of each stream in each year, the window used for the thermal index (Section A3), and the observed spawning period.

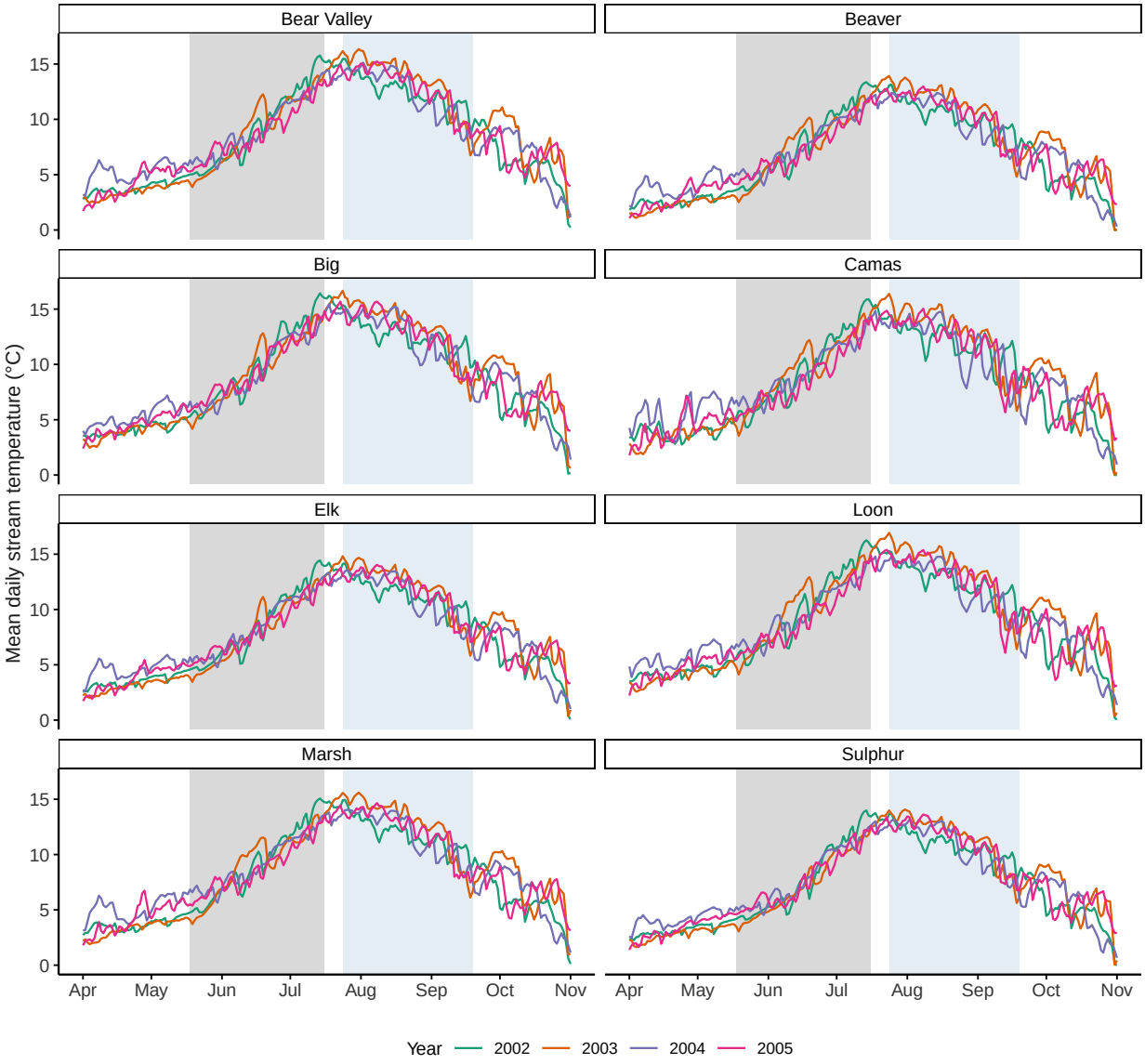


Figure A1.3: Modeled mean daily stream temperature across surveyed reaches, by stream and year (Siegel et al. 2023). Grey band: window of the thermal index (60-d mean, 18 May–16 Jul). Blue band: range of observed spawn dates.

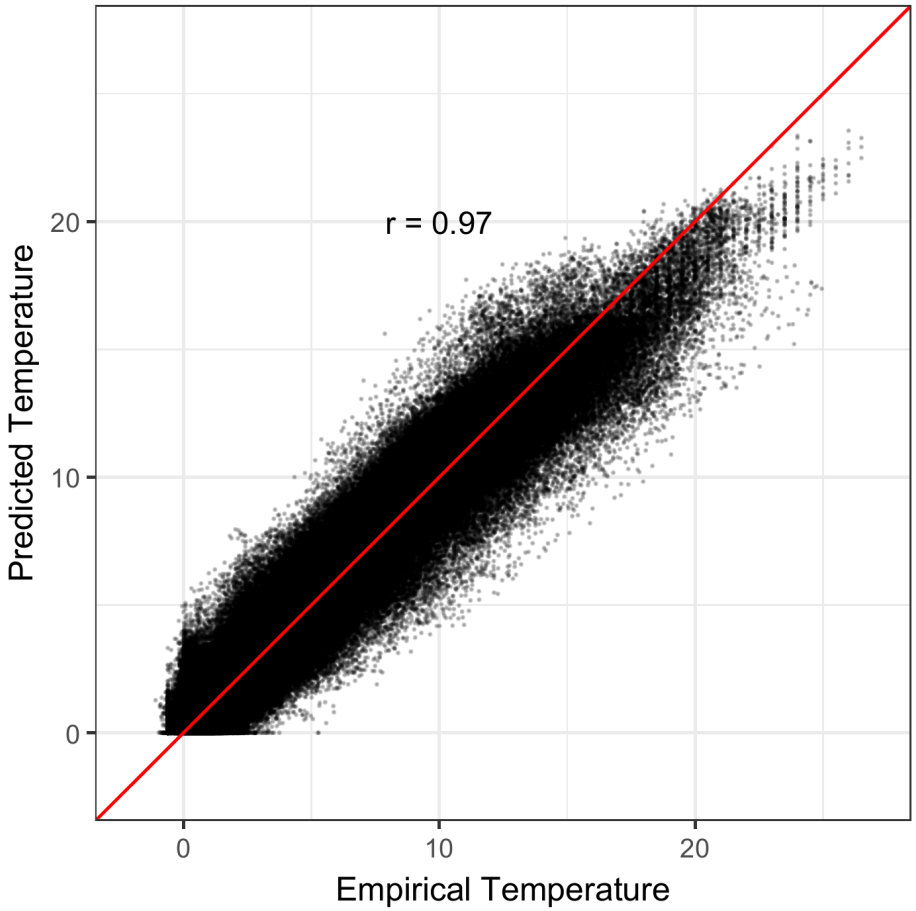


Figure A1.4: Comparison of modeled daily stream temperatures (Siegel et al. 2023) with empirical logger observations (Isaak et al.).

A2 Why the antecedent temperature window was replaced

In the original submission, each redd's thermal covariate was the mean modeled temperature over the 90 days ending on that redd's own spawn date ("temp_90"). Because the window is anchored to the response, spawn date enters both sides of the model: any seasonal trend in stream temperature produces an association between spawn date and temp_90 even if spawning is unrelated to temperature. This appendix quantifies that artefact. The analyses are in `R/07_window-artefact-diagnostics.R`.

A2.1 Seasonal context

Stream temperature across surveyed reaches peaked in late July, and no redd was completed before the peak (Figure A2.1). A 90-day window ending in August or September therefore spans cold late-spring days at its start and warm summer days at its end. Moving the end date later drops cold days and adds warm ones, so a later spawn date mechanically yields a warmer temp_90.

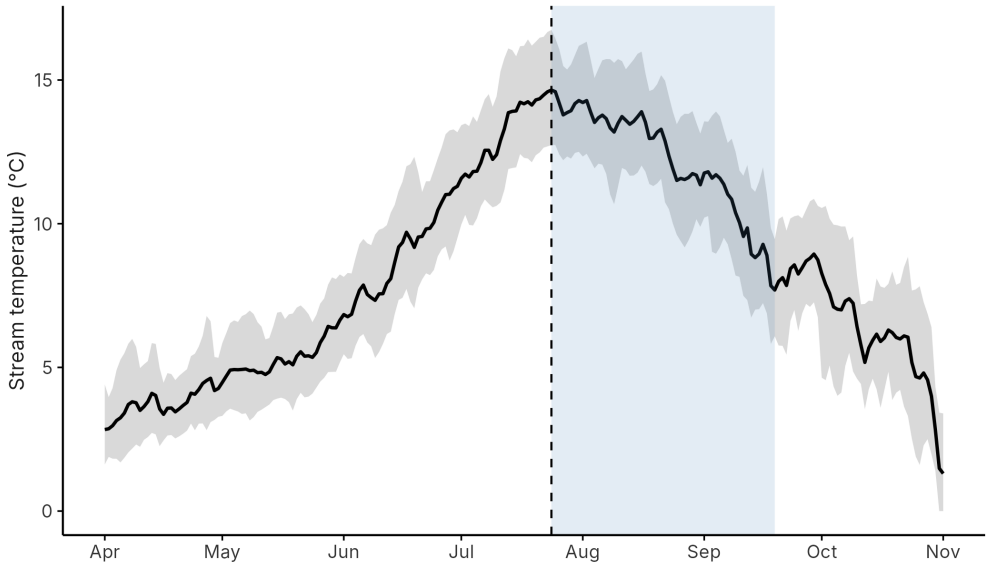


Figure A2.1: Network-mean modeled daily stream temperature for surveyed reaches, 2002–2005 (grey band: 10th–90th percentiles). Shaded: range of observed spawn dates; dashed line: thermal peak.

A2.2 Within a reach-year, the window re-codes the date

Redds in the same reach and year share the same stream; they differ only in spawn date. Within reach-years, temp_90 rose 0.73 °C for every 10 days of later spawning and was almost perfectly correlated with spawn date (r = 0.96; Table A2.1). The direction of the artefact depends on window length: a 30-d window ending in August sits on the summer peak and becomes *cooler* as it moves later, whereas the 90-d window becomes warmer. This explains why the 90-d window was the best predictor in the original screening, and why biologically more plausible shorter windows were not: only the 90-d artefact reinforces the spatial pattern.

Table A2.1: Change in date-anchored antecedent temperature with spawn date among redds in the same reach and year.

Window	Change per 10 d later (°C)	r with date, within reach-year	r with date, raw	Variance within reach-year
30-d antecedent mean	-0.58	-0.82	0.08	7%
60-d antecedent mean	+0.34	0.69	0.65	3%
90-d antecedent mean	+0.73	0.96	0.82	5%

Table A2.2: Observed fit statistics compared with 200 shuffled-date null datasets.

	R^2 (quadratic)	R^2 (+ stream + year)	Within reach-year slope (d per °C)	Mixed-model residual SD (d)
Observed	0.71	0.84	12.6	2.03
Null, mean	0.30	0.46	13.5	2.40
Null, 2.5%	0.28	0.44	13.4	2.35
Null, 97.5%	0.32	0.48	13.6	2.46

A2.3 A null model with no biology

To ask how much of the original fit the artefact alone could produce, we shuffled spawn dates among redds within each year (200 permutations), recomputed temp_90 at each redd's own reach for its new date, and refit the models. Shuffling destroys any real link between a place's temperature and its spawn timing but keeps the window construction intact.

Random dates still produced $R^2 = 0.30$ for spawn date on temp_90 (quadratic; observed 0.71), and a within-reach slope of 13.5 d per °C, slightly steeper than observed (12.6; Table A2.2, Figure A2.2). The mixed model fit to random dates predicted them with a residual SD of 2.4 d, against 2.0 d for the real data. The within-reach temperature response in the original model is therefore indistinguishable from the artefact; the real signal lies in temperature differences among reaches and years.

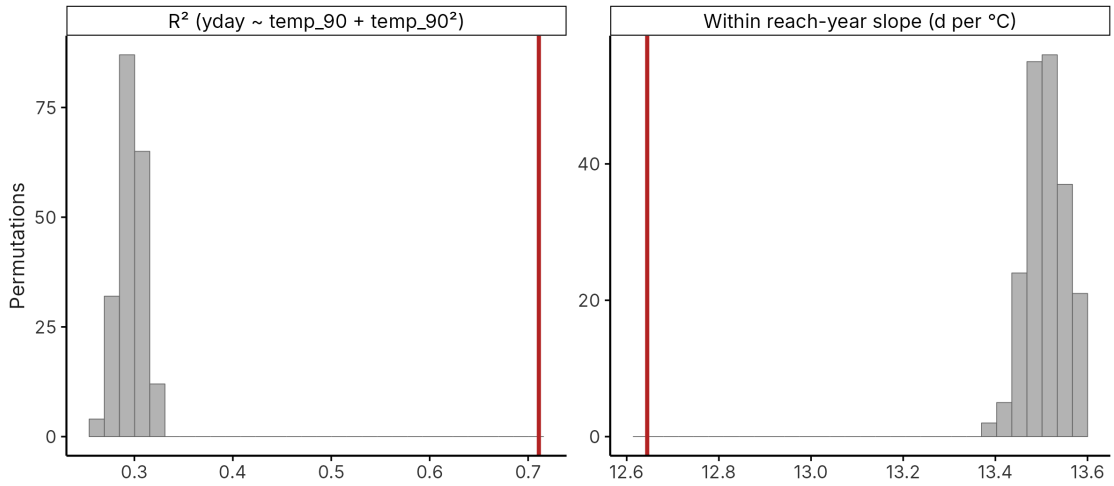


Figure A2.2: Distribution of fit statistics from 200 shuffled-date null datasets (grey) and the observed value (red).

A2.4 Reach-specific slopes reflect thermograph shape

If temp_90 rises at a rate of b °C per day within a reach-year, a regression of spawn date on temp_90 within that reach has slope $1/b$ days per °C regardless of biology. For the 37 reaches with at least 15 redds, the observed within-reach slope tracked this artefact-only expectation ($r = 0.75$; medians 13.1 and 13.1 d per °C; Figure A2.3). The reach-specific random slopes reported in the original submission ("thermal sensitivity") therefore largely describe how quickly each reach's 90-day mean warms through August, not how fish respond to temperature. The new analysis uses a thermal index that is fixed in calendar time (Section A3).

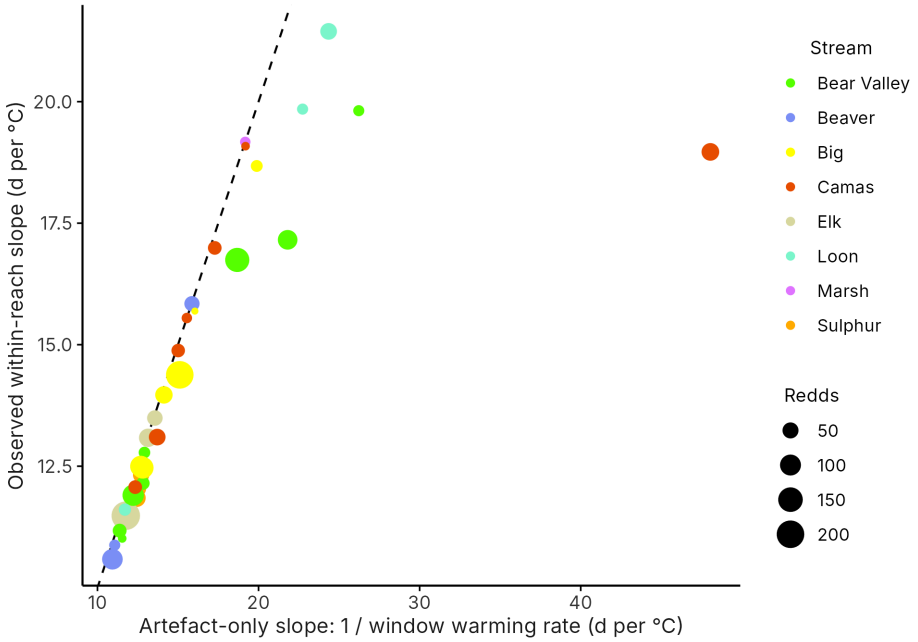


Figure A2.3: Observed within-reach slope of spawn date on temp_90 versus the slope expected from window construction alone (1 / rate at which the window warms), for reaches with at least 15 redds. Dashed line: 1:1.

A3 Choosing a fixed-date thermal index

We replaced temp_90 with reach-year thermal indices computed over the same calendar window for every redd in a reach-year, so the index cannot depend on spawn date. The analyses are in `R/08_fixed-date-index.R`.

A3.1 Pre-spawn indices

We compared 32 candidates: mean temperature over 30, 60 or 90 days ending on ten dates from 1 July to 15 August, plus the mean for 1 July–31 August and the maximum 7-day mean (MWAT) for June–September. Each was fit with the same model, spawn date ~ index + stream + year, with random intercepts for reach and reach-year (the reach-year term is explained in Section A5).

The data could not distinguish among the candidates: all fell within 23 AIC units of the best (85 without the reach-year term), and the estimated effect ranged from 3.8 to 6.1 days per °C (Table A3.1, Figure A3.1). These indices are strongly correlated and rank reaches and years in nearly the same order. We therefore chose the index on biological grounds: the mean of the 60 days ending 16 July (18 May–16 July), which ends before the first observed redd (24 July) and so contains no temperatures experienced after any fish spawned. Because most adults arrive after this window, we interpret the index as a signature of each reach's thermal regime, not as the temperature individual fish experienced. Over 1990–2021 the index ranked reaches in almost the same order every year (Section A7), which supports this interpretation.

Table A3.1: Candidate fixed-date thermal indices ranked by AIC. Slope is the linear effect (SE) from spawn date ~ index + stream + year with reach and reach-year random intercepts. Pre-spawn: window ends before the first observed redd.

Index	Pre-spawn	ΔAIC	ΔAIC (no reach-year term)	Slope (d per °C)	R^2 marginal	r (reach-year means)
mean Jul1-Aug31	no	0.0	49.3	4.16 (0.27)	0.546	0.58
30-d mean ending 15Aug	no	1.3	65.1	3.99 (0.26)	0.546	0.55
MWAT Jun-Sep	no	2.5	22.7	3.76 (0.25)	0.544	0.54
90-d mean ending 15Aug	no	2.9	11.8	4.71 (0.31)	0.540	0.64
60-d mean ending 15Aug	no	3.4	39.7	4.21 (0.28)	0.541	0.59
90-d mean ending 10Aug	no	3.8	14.0	4.82 (0.32)	0.538	0.65
60-d mean ending 10Aug	no	4.4	32.7	4.28 (0.28)	0.541	0.59
30-d mean ending 10Aug	no	5.1	71.6	3.93 (0.26)	0.542	0.58
30-d mean ending 05Aug	no	5.7	72.2	3.94 (0.26)	0.541	0.57
90-d mean ending 05Aug	no	5.8	18.1	4.94 (0.33)	0.535	0.65
60-d mean ending 05Aug	no	8.1	34.0	4.37 (0.30)	0.538	0.58
90-d mean ending 31Jul	no	8.4	23.6	5.07 (0.35)	0.532	0.65
30-d mean ending 31Jul	no	8.9	85.3	3.96 (0.27)	0.537	0.56
30-d mean ending 26Jul	no	10.1	84.2	4.02 (0.27)	0.534	0.52
30-d mean ending 11Jul	yes	10.2	5.8	4.56 (0.33)	0.533	0.57
90-d mean ending 26Jul	no	10.2	27.3	5.18 (0.36)	0.530	0.65
60-d mean ending 31Jul	no	10.3	29.0	4.49 (0.31)	0.535	0.59
90-d mean ending 21Jul	yes	10.4	22.0	5.37 (0.38)	0.529	0.65
60-d mean ending 16Jul	yes	10.5	0.0	5.02 (0.36)	0.530	0.65
30-d mean ending 16Jul	yes	10.6	22.8	4.37 (0.31)	0.530	0.54
90-d mean ending 16Jul	yes	10.6	15.5	5.58 (0.40)	0.529	0.65
60-d mean ending 11Jul	yes	10.8	0.3	5.23 (0.38)	0.529	0.66
60-d mean ending 21Jul	yes	11.3	15.5	4.79 (0.34)	0.531	0.63
90-d mean ending 11Jul	yes	11.4	11.3	5.78 (0.42)	0.527	0.64
60-d mean ending 26Jul	no	12.1	30.2	4.59 (0.33)	0.531	0.61
90-d mean ending 06Jul	yes	12.1	2.7	5.95 (0.43)	0.525	0.63
30-d mean ending 21Jul	yes	12.3	75.8	4.16 (0.28)	0.530	0.52
90-d mean ending 01Jul	yes	13.4	0.2	6.12 (0.45)	0.523	0.62
60-d mean ending 06Jul	yes	14.0	3.9	5.42 (0.40)	0.524	0.66
60-d mean ending 01Jul	yes	17.3	8.8	5.58 (0.42)	0.520	0.65
30-d mean ending 06Jul	yes	19.7	18.3	4.67 (0.36)	0.525	0.58
30-d mean ending 01Jul	yes	23.2	3.3	4.81 (0.38)	0.520	0.60

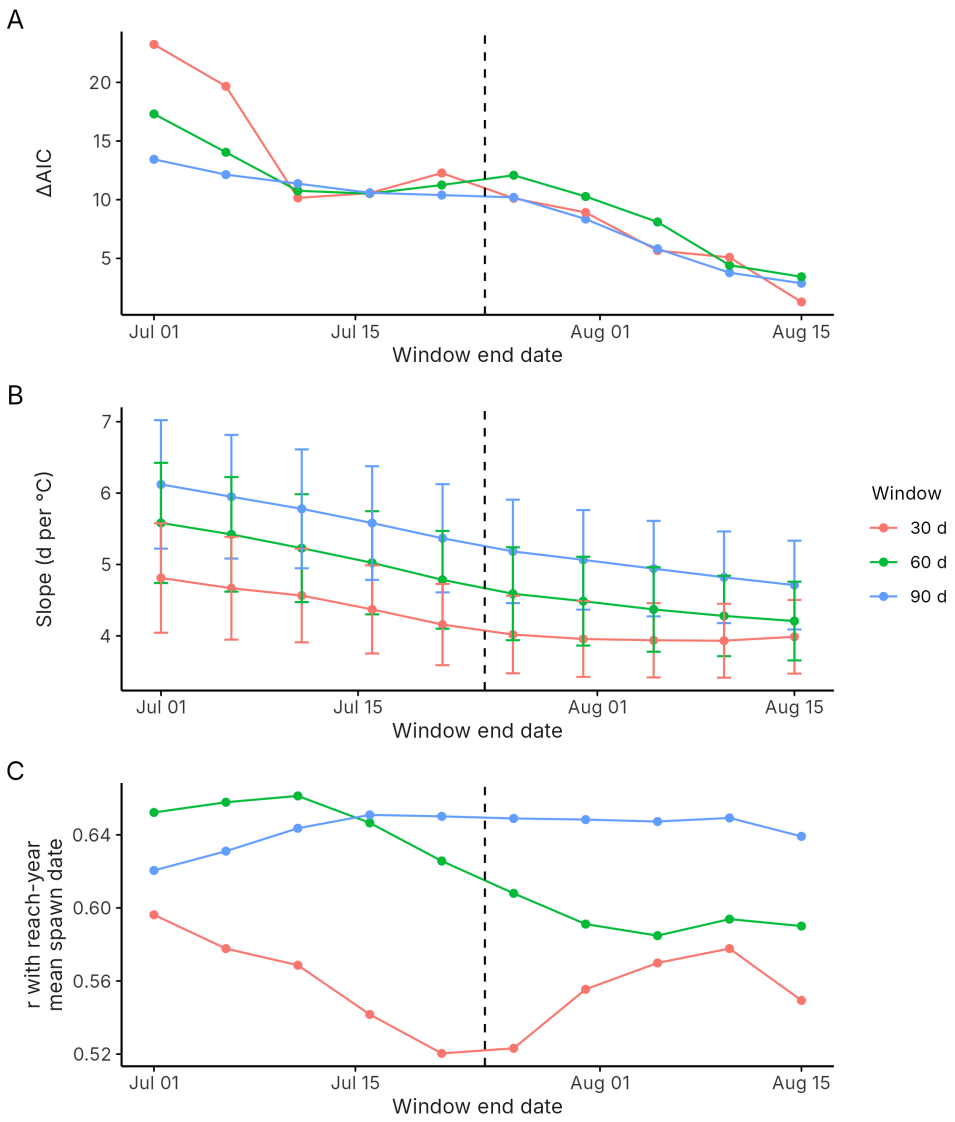


Figure A3.1: Fit (ΔAIC), effect size (± 2 SE) and correlation with reach-year mean spawn date for fixed-date indices of 30, 60 and 90 days, by window end date. Dashed line: first observed redd.

A3.2 Post-spawn (incubation) indices

Salmon may time spawning so that fry emerge in a favourable spring window, so that reaches with colder incubation spawn earlier (Heggberget 1988; Brannon et al. 2004). The original submission tested post-spawn windows that began on each redd's own spawn date, which carries the mirror image of the artefact in Section A2. We instead computed fixed-date reach-year means for fall (1 Sep–30 Nov), winter (1 Dec–28 Feb), spring (1 Mar–30 Apr) and the whole incubation period (1 Sep–30 Apr), and the number of days to emergence for a fixed reference spawn date (1 Sep or 15 Aug), using the Beacham and Murray (1990) Chinook emergence model (model 2) as implemented in hatchR (Sparks et al. 2026). For a 1 September spawn, predicted emergence ranged from 26 Mar to 09 Jul among reach-years.

On their own, all post-spawn indices predicted spawn timing in the direction the hypothesis predicts (warmer incubation, later spawning; Table A3.2). For example, spawn date shifted 0.24 days earlier for each additional day of predicted incubation, offsetting roughly a quarter of the among-reach difference in development time. However, post-spawn indices were strongly correlated with the pre-spawn index (r up to 0.87). When added

Table A3.2: Post-spawn (incubation) indices. Alone: spawn date ~ index + stream + year. Added: index added to the final model (Section A5), with all streams and without Big Creek. Slopes are days per °C, or days per day of predicted incubation for emergence indices. Negative ΔAIC = better fit.

Index	r with pre-spawn index	Alone: slope (SE); ΔAIC vs pre-spawn	Added to final: slope (SE); ΔAIC	Added, no Big Creek: slope (SE); ΔAIC
Fall mean, 1 Sep–30 Nov	0.86	7.35 (0.47); -14.6	11.25 (2.13); -22.8	0.63 (3.17); +2.0
Winter mean, 1 Dec–28 Feb	0.70	13.58 (1.70); +64.6	1.13 (1.99); +1.7	-1.88 (1.89); +1.0
Spring mean, 1 Mar–30 Apr	0.72	7.82 (0.55); +1.2	4.32 (1.57); -5.4	2.31 (1.87); +0.5
Incubation mean, 1 Sep–30 Apr	0.87	10.09 (0.67); -4.5	7.78 (2.35); -8.2	-0.04 (2.79); +2.0
Days to emergence if spawned 1 Sep	-0.86	-0.24 (0.02); +2.4	-0.15 (0.06); -4.6	0.04 (0.06); +1.6
Days to emergence if spawned 15 Aug	-0.85	-0.13 (0.01); +4.3	-0.07 (0.03); -5.2	0.02 (0.03); +1.7

to the final model they improved fit only because of Big Creek; without Big Creek, none improved fit and their effects were near zero (Table A3.2). The data are therefore consistent with incubation matching but cannot separate it from the summer thermal regime of a reach (see also Section A4).

A4 Other covariates and the Big Creek gradient

Reach elevation and reach slope (NHDPlus Version 2) and discharge were considered as additional covariates. Results are in `R/09_refit-fixed-date-model.R` (F5) and Table 1 of the main text.

A4.1 Elevation

Elevation is fixed for each reach, so it can only compete with the between-reach component of temperature ($\bar{T}$; Section A5). Added to the final model, elevation improved fit by 40 AIC units: higher reaches spawned 2.8 days earlier per 100 m (SE 0.4), and the between-reach temperature effect fell from 4.9 to 1.6 days per °C (Table A4.1). The within-reach temperature effect was unchanged (6.4 days per °C).

This effect depends on one stream. Surveyed reaches in Big Creek span 647 m of elevation, compared with 22–303 m elsewhere (Table A1.1). Without Big Creek, elevation had no detectable effect (-0.6 days per 100 m, SE 0.6) and did not improve fit (ΔAIC = +0.9). Allowing separate elevation slopes gave -3.0 (SE 0.4) days per 100 m in Big Creek and -1.1 (SE 0.6) elsewhere (Figure A4.1). We therefore did not include elevation in the final model.

Within Big Creek, upper reaches spawned earlier than their summer temperatures predict, and lower reaches later. Several explanations fit this single gradient and cannot be separated with these data: run-timing structure (summer-run fish spawn in lower Big Creek; Copeland et al. 2019), migration distance or arrival order, and colder fall and winter incubation conditions in upper reaches (post-spawn indices behaved the same way as elevation; Section A3).

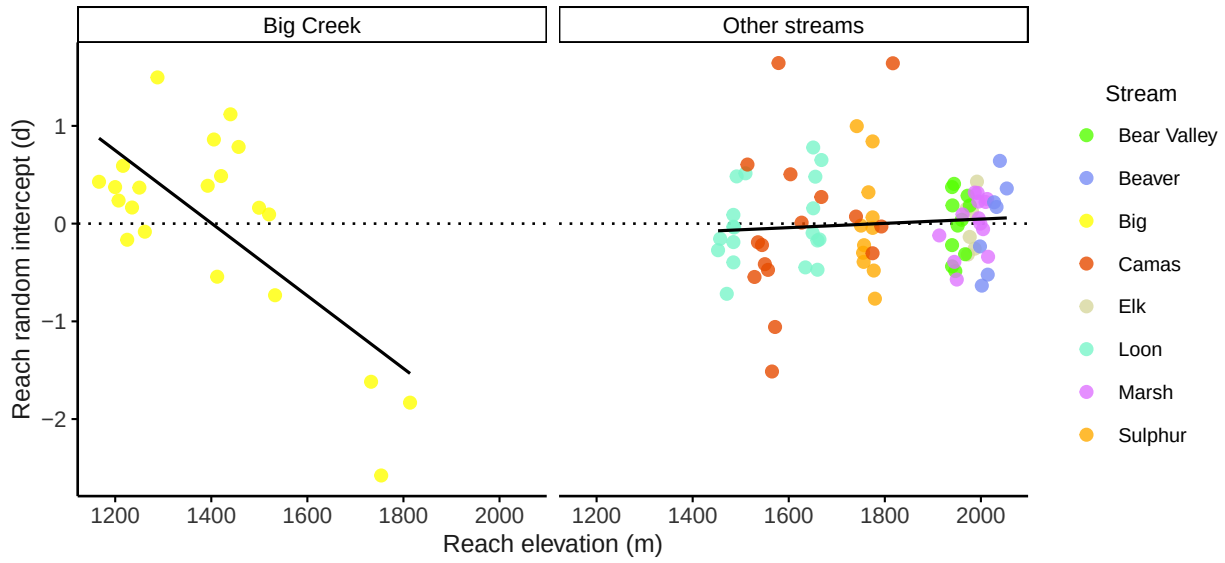


Figure A4.1: Reach random intercepts from the final model (no elevation) against reach elevation, for Big Creek and the other streams. Lines: linear fits. Positive intercepts = later spawning than temperature, stream and year predict.

Table A4.1: Elevation added to the final model, with all streams and without Big Creek. Temperature effects ($\bar{T}$, reach mean; $T - \bar{T}$, reach-year anomaly) in days per °C; SE in parentheses.

Data	$\bar{T}$, no elevation	$\bar{T}$, with elevation	$T - \bar{T}$, with elevation	Elevation (d per 100 m)	ΔAIC adding elevation
All streams	4.91 (0.37)	1.56 (0.58)	6.35 (1.48)	-2.80 (0.40)	-40.2
Without Big Creek	3.38 (0.39)	2.85 (0.64)	6.53 (1.55)	-0.63 (0.61)	+0.9

A4.2 Reach slope

Reach slope ranged narrowly (0.0–4.8%) and did not improve the final model (ΔAIC = +1.9; Table 1 of the main text).

A4.3 Discharge

Discharge was available only from one gauge downstream of all study reaches (USGS 13309220), so it varies among years but not among reaches, and it is confounded with the year effect. Discharge summaries in the original submission were also computed over windows anchored to each redd's spawn date and so share the artefact in Section A2. We did not retain discharge.

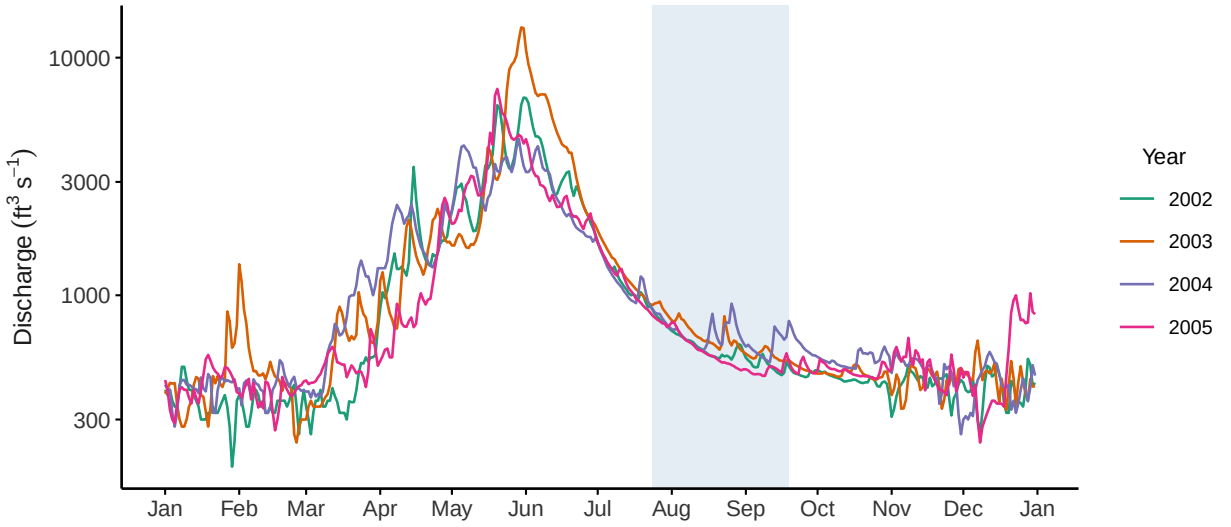


Figure A4.2: Mean daily discharge at the Middle Fork Salmon River gauge near Middle Fork Lodge (USGS 13309220), 2002–2005.

A5 Model structure, selection and diagnostics

A5.1 Model structure

Spawn day of year for redd i in reach j and year t was modelled as

$$y_{ijt} = \beta_0 + \beta_1 \bar{T}_j + \beta_2 (T_{jt} - \bar{T}_j) + \text{stream}_{s(j)} + \text{year}_t + u_j + v_{jt} + \varepsilon_{ijt},$$

where T_{jt} is the thermal index for reach j in year t (Section A3), $\bar{T}_j$ is the mean of T_{jt} over the years reach j was surveyed, $u_j \sim N(0, \sigma^2_{\text{reach}})$ and $v_{jt} \sim N(0, \sigma^2_{\text{reach-year}})$ are random intercepts for reach and reach-year, and $\varepsilon_{ijt} \sim N(0, \sigma^2)$. Models were fit with lme4 (Bates et al. 2015); comparisons used maximum likelihood and final estimates used REML.

Splitting temperature into two parts. Writing the thermal index as $\bar{T}_j$ plus the anomaly $T_{jt} - \bar{T}_j$ (within-subject centring; van de Pol and Wright 2009) separates two contrasts that a single temperature term mixes. β_1 compares warmer and cooler *reaches*; it can be confounded with anything else that differs among places (elevation, migration distance, run timing). β_2 compares the *same reach* in warmer and cooler years; fixed reach characteristics cannot produce it, and with year fixed effects in the model it is estimated from how much a reach's temperature departed from its own average beyond the network-wide shift that year. The split does not change model fit (Table A5.1).

Why a reach-year random intercept. The thermal index has one value per reach-year, so redds in the same reach-year share it, and they also share unmeasured conditions (arrival timing, flows, the few females that spawned there). Without a reach-year term, a reach-year with 100 redds counts as 100 independent confirmations of one temperature–date pair. In a model without this term, residuals clustered strongly by reach-year (intraclass correlation 0.22), and adding the term improved AIC by 436 units. Without it, random slopes for temperature also appeared to be supported; with it, they were not (singular fit).

A5.2 Model comparison

A5.3 Shape of the temperature effect

A natural spline on $\bar{T}$ improved AIC slightly (-3.3; Table A5.1). The fitted curve departs little from the straight line over the range of the data (Figure A5.1), so we retained the linear term for interpretability. The plateau at warm sites reported in the original submission is not present with the fixed-date index.

Table A5.1: Model comparison (maximum likelihood; n = 3,016 redds). Within each group, ΔAIC is relative to the first row. T: reach-year thermal index; $\bar{T}$: reach mean; $T - \bar{T}$: reach-year anomaly. R^2 conditional is undefined for singular fits.

Model	k	ΔAIC	RMSE (d)	R^2 marginal	R^2 conditional	Notes
Random-effect structure (fixed effects as in the final model)						
Reach + reach-year intercepts	16	0.0	4.19	0.535	0.705	Final model
Reach intercept only	15	435.6	4.80	0.560	0.635	
Reach + reach-year intercepts, random slopes for (T − $\bar{T}$)	18	3.7	4.19	0.646	–	Singular fit; not supported
Temperature terms (with stream + year)						
$\bar{T}$ + (T − $\bar{T}$)	16	0.0	4.19	0.535	0.705	Final model
T (not split)	15	-0.4	4.19	0.530	0.702	
$\bar{T}$ only	15	16.6	4.19	0.504	0.695	
(T − $\bar{T}$) only	15	94.4	4.18	0.378	0.724	
No temperature	14	109.9	4.17	0.343	0.717	
Other fixed effects (relative to the final model)						
$\bar{T}$ + (T − $\bar{T}$) + stream + year	16	0.0	4.19	0.535	0.705	Final model
− stream	9	65.0	4.18	0.355	0.691	
− year	13	22.7	4.19	0.498	0.699	
+ elevation	17	-40.2	4.20	0.670	–	Driven by Big Creek; not supported without it (Appendix)
+ reach slope	17	1.9	4.19	0.535	0.705	
Shape of the between-reach temperature effect						
Linear $\bar{T}$	16	0.0	4.19	0.535	0.705	Final model
Natural spline on $\bar{T}$ (df = 3)	18	-3.3	4.19	0.546	0.706	Slight curvature; linear retained (Appendix)

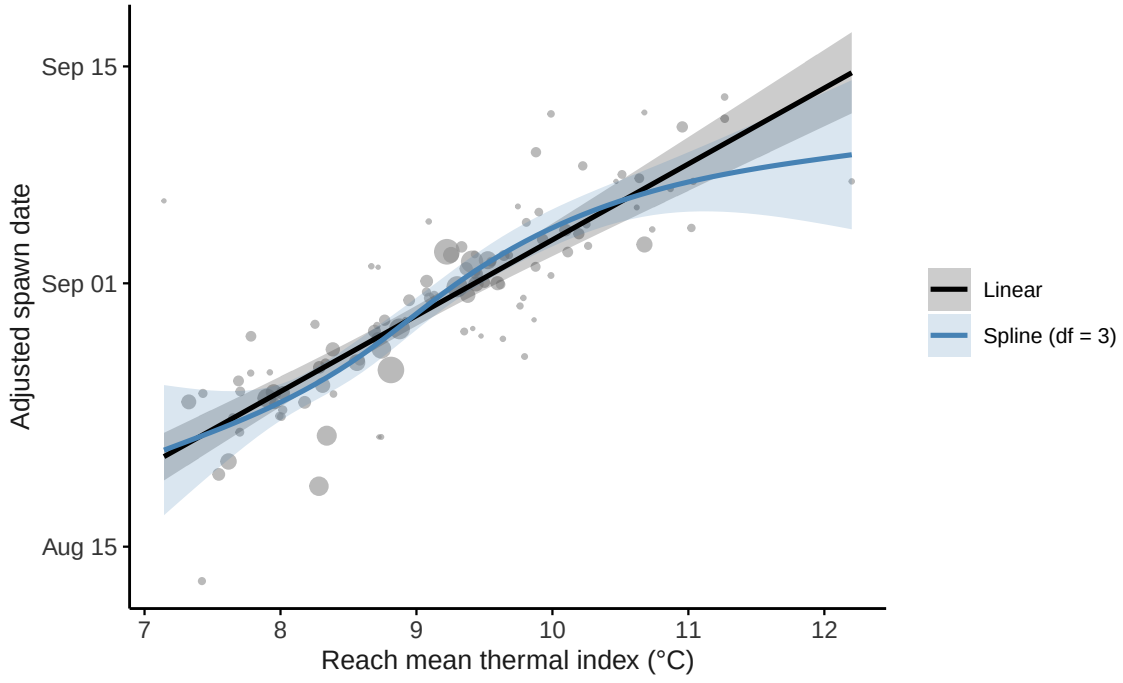


Figure A5.1: Between-reach temperature effect fitted as linear (black) and as a natural spline with 3 df (blue), with 95% CIs. Points: reach mean spawn dates adjusted for stream, year and the reach-year anomaly.

A5.4 Diagnostics for the final model

Residuals showed no trend with fitted values, stream or year, but had heavier tails than a normal distribution, reflecting a small number of unusually early or late redds (Figure A5.2). With 3,016 redds and 263 reach-years, fixed-effect estimates are robust to this departure; random-effect distributions were close to normal. The residual spread reflects the spread of spawn dates within reach-years (residual SD 4.3 d), which the model does not attempt to explain; predicted reach-year means track observed means closely (Figure A5.3).

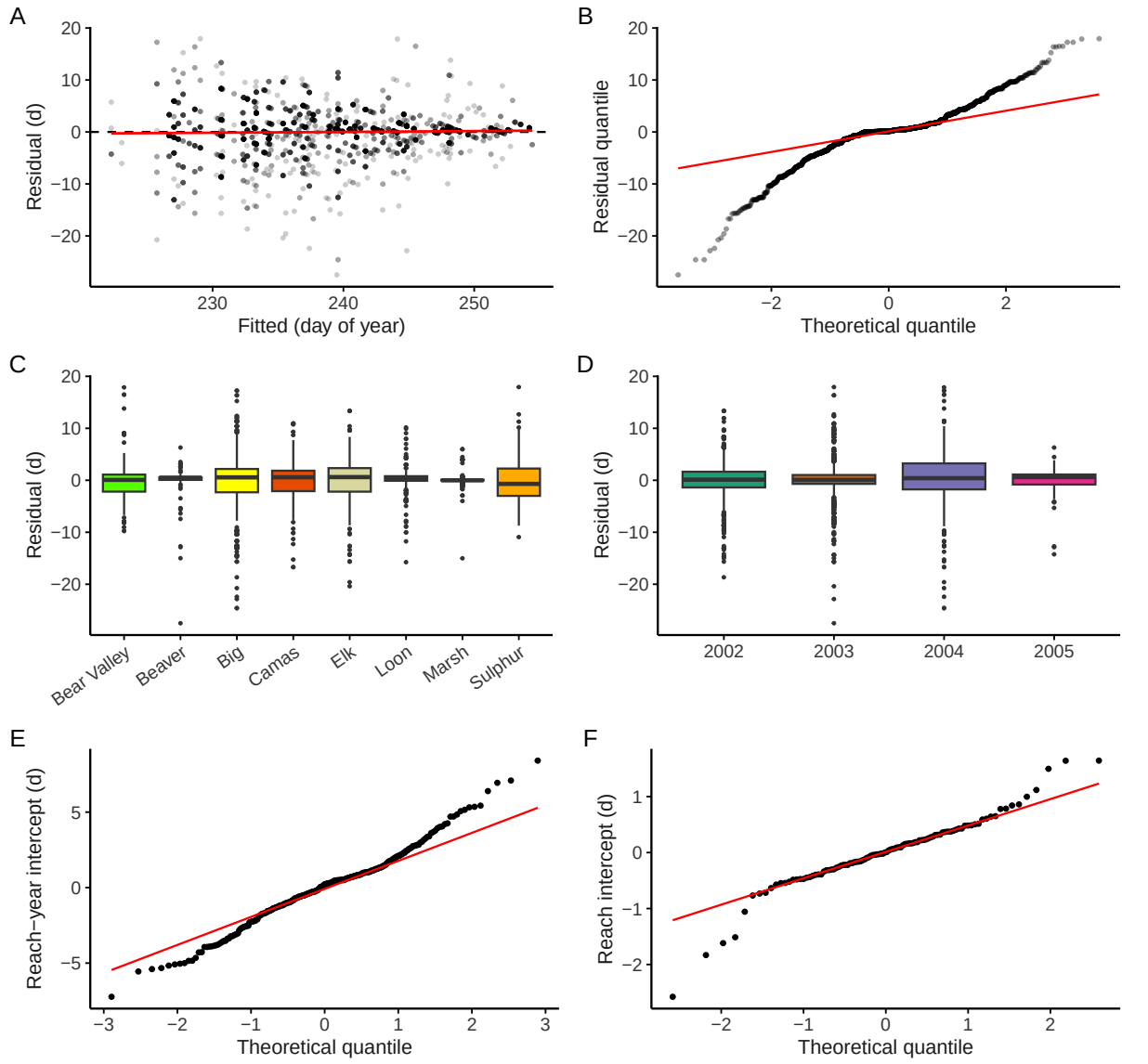


Figure A5.2: Residual diagnostics for the final model. (A) Residuals vs fitted values. (B) Normal Q–Q plot of residuals. (C) Residuals by stream. (D) Residuals by year. (E) Normal Q–Q plot of reach-year random intercepts. (F) Normal Q–Q plot of reach random intercepts.

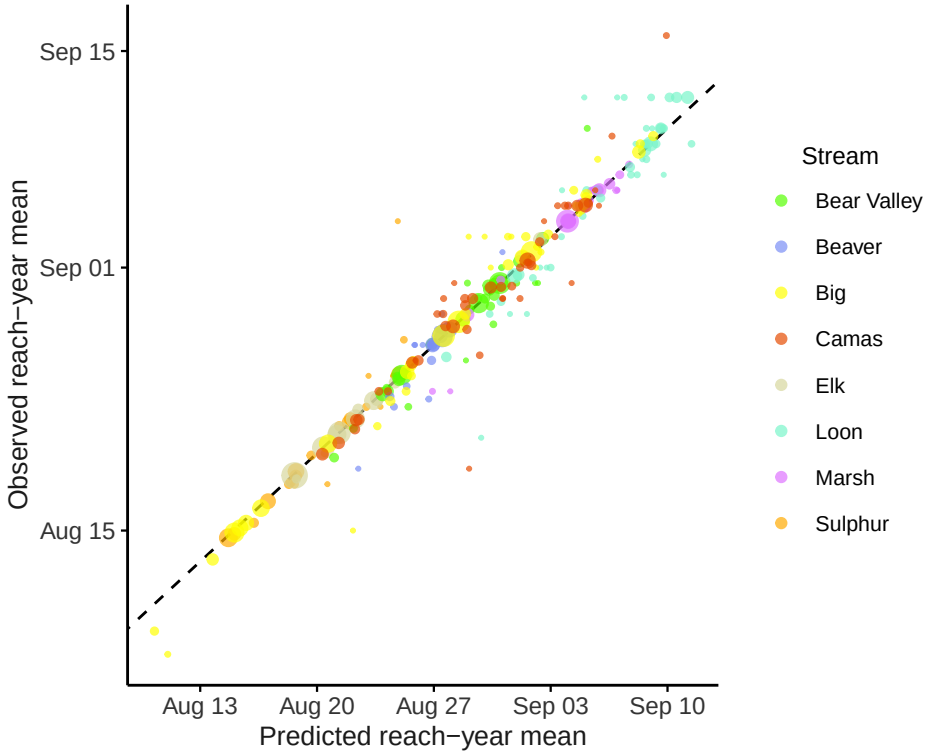


Figure A5.3: Observed vs predicted mean spawn date for each reach-year (predictions include random effects). Dashed line: 1:1. Point size: number of redds.

A6 Final model results

A6.1 Parameter estimates

Spawning was 4.9 days later per °C between reaches and 7.0 days later per °C within reaches between years. The within-reach estimate rests on anomalies of about ±0.5 °C (Fig. 4 of the main text), so it corresponds to shifts of roughly ±3 days (see Section A7 for how these anomalies compare with the 1990–2021 record). Once temperature, stream and year were accounted for, persistent differences among reaches were small (reach SD 1.4 d) compared with year-specific shifts within reaches (reach-year SD 3.1 d) and the spread of individual redds (residual SD 4.3 d).

A6.2 Stream and year means

Marginal means are predicted spawn dates at the average reach temperature, averaged over the other factor (emmeans; Lenth 2024). They differ from observed means where a stream's or year's temperatures differ from average (Fig. 5 of the main text).

A6.3 Reach effects and position in the network

Reach random intercepts showed no consistent upstream–downstream pattern within streams: the pooled within-stream slope on log drainage area was 0.12 days per log unit (SE 0.07; Figure A6.1). Big Creek was the exception, with downstream reaches spawning later than predicted; this is the same gradient that drives the elevation result in Section A4.

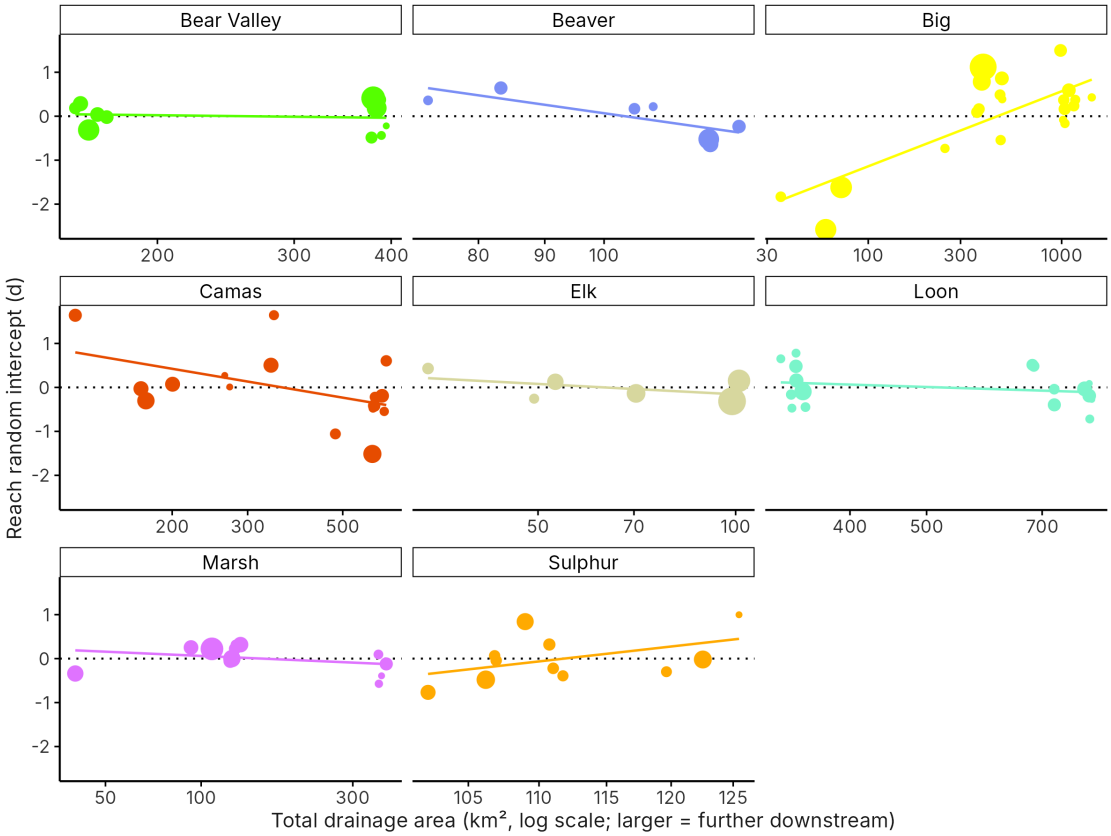


Figure A6.1: Reach random intercepts from the final model against total drainage area (log scale; larger = further downstream), by stream. Point size: number of redds; lines: linear fits.

Table A6.1: Final model estimates (REML). Reference levels: Bear Valley Creek, 2002. The reach mean thermal index is centred at the average reach, so the intercept is the predicted spawn day of year for Bear Valley Creek in 2002 at average reach temperature.

Parameter	Estimate	95% CI	p
Fixed effects			
Intercept (Bear Valley, 2002, average reach)	238.31	[236.50, 240.12]	<0.001
Reach mean thermal index, $\bar{T}$ (d per °C)	4.90	[4.12, 5.67]	<0.001
Reach-year anomaly, T − $\bar{T}$ (d per °C)	7.01	[3.85, 10.18]	<0.001
Stream: Beaver	5.13	[2.47, 7.80]	<0.001
Stream: Big	-0.99	[-3.01, 1.04]	0.339
Stream: Camas	0.87	[-1.10, 2.84]	0.388
Stream: Elk	-1.98	[-4.72, 0.75]	0.155
Stream: Loon	5.96	[3.89, 8.02]	<0.001
Stream: Marsh	6.45	[4.32, 8.58]	<0.001
Stream: Sulphur	0.72	[-1.84, 3.27]	0.583
Year: 2003	1.50	[0.21, 2.78]	0.022
Year: 2004	1.63	[0.18, 3.07]	0.028
Year: 2005	6.94	[4.43, 9.45]	<0.001
Random effects			
SD reach-year intercept (d)	3.14		
SD reach (COMID) intercept (d)	1.36		
SD residual (d)	4.33		
N reach-years	263		
N reaches (COMID)	104		
N redds	3016		
Model performance			
R^2 marginal	0.53		
R^2 conditional	0.71		
ICC (adjusted)	0.38		
RMSE (days)	4.19		

Table A6.2: Observed and temperature-adjusted (marginal) mean spawn dates by stream and year.

	Redds	Observed mean	Adjusted mean	95% CI	Day of year (SE)
Stream					
Bear Valley	479	29 Aug	28 Aug	26 Aug – 29 Aug	239.8 (0.74)
Beaver	203	27 Aug	02 Sep	31 Aug – 04 Sep	244.9 (1.12)
Big	647	26 Aug	27 Aug	25 Aug – 28 Aug	238.8 (0.76)
Camas	323	29 Aug	29 Aug	27 Aug – 30 Aug	240.7 (0.71)
Elk	471	22 Aug	26 Aug	23 Aug – 28 Aug	237.8 (1.19)
Loon	242	06 Sep	03 Sep	01 Sep – 04 Sep	245.8 (0.79)
Marsh	368	04 Sep	03 Sep	02 Sep – 05 Sep	246.2 (0.82)
Sulphur	283	19 Aug	29 Aug	26 Aug – 31 Aug	240.5 (1.06)
Year					
2002	1187	27 Aug	27 Aug	26 Aug – 28 Aug	239.3 (0.50)
2003	1217	28 Aug	29 Aug	28 Aug – 30 Aug	240.8 (0.47)
2004	404	28 Aug	29 Aug	28 Aug – 30 Aug	240.9 (0.57)
2005	208	02 Sep	03 Sep	01 Sep – 05 Sep	246.2 (1.04)

Table A6.3: Pairwise differences in adjusted mean spawn date between streams and between years (Tukey-adjusted).

	Difference (d)	SE	z	Tukey p
Stream				
Bear Valley − Beaver	-5.13	1.36	-3.77	0.004
Bear Valley − Big	0.99	1.03	0.96	0.980
Bear Valley − Camas	-0.87	1.01	-0.86	0.989
Bear Valley − Elk	1.98	1.39	1.42	0.847
Bear Valley − Loon	-5.96	1.06	-5.64	<0.001
Bear Valley − Marsh	-6.45	1.09	-5.94	<0.001
Bear Valley − Sulphur	-0.72	1.30	-0.55	0.999
Beaver − Big	6.12	1.39	4.39	<0.001
Beaver − Camas	4.26	1.37	3.12	0.039
Beaver − Elk	7.12	1.59	4.49	<0.001
Beaver − Loon	-0.83	1.46	-0.56	0.999
Beaver − Marsh	-1.32	1.42	-0.93	0.983
Beaver − Sulphur	4.41	1.36	3.26	0.025
Big − Camas	-1.86	0.96	-1.93	0.527
Big − Elk	1.00	1.38	0.72	0.996
Big − Loon	-6.94	0.99	-7.04	<0.001
Big − Marsh	-7.44	1.02	-7.26	<0.001
Big − Sulphur	-1.70	1.31	-1.30	0.899
Camas − Elk	2.85	1.37	2.09	0.424
Camas − Loon	-5.09	0.99	-5.14	<0.001
Camas − Marsh	-5.58	1.03	-5.40	<0.001
Camas − Sulphur	0.15	1.30	0.12	1.000
Elk − Loon	-7.94	1.42	-5.58	<0.001
Elk − Marsh	-8.44	1.42	-5.95	<0.001
Elk − Sulphur	-2.70	1.50	-1.80	0.621
Loon − Marsh	-0.50	1.07	-0.47	1.000
Loon − Sulphur	5.24	1.41	3.71	0.005
Marsh − Sulphur	5.74	1.34	4.28	<0.001
Year				
2002 − 2003	-1.50	0.66	-2.29	0.101
2002 − 2004	-1.63	0.74	-2.20	0.122
2002 − 2005	-6.94	1.28	-5.42	<0.001
2003 − 2004	-0.13	0.67	-0.19	0.997
2003 − 2005	-5.44	1.09	-5.01	<0.001
2004 − 2005	-5.31	1.09	-4.88	<0.001

A7 The study years in the long-term thermal record

Siegel et al. (2023) predict daily temperatures for every year from 1990 to 2021, so we computed the thermal index (Section A3) for every surveyed reach in every year of that record (`R/12_long-term-thermal-context.R`). This shows how the four study years compare with the longer record. It also shows how much of the variation in the index is fixed among reaches and how much changes between years.

A7.1 Were 2002–2005 typical years?

Network-wide, the study years were average to slightly warm, and very similar to one another (Figure A7.1). The mean index across surveyed reaches ranged from 8.9 °C (2005) to 9.4 °C (2002), against a 1990–2021 mean of 8.9 °C. The study years ranked 16th to 22th coldest of 32 years. Over the full record the network mean ranged from 6.8 to 11.3 °C, so the four study years span about 12% of the between-year range in the record.

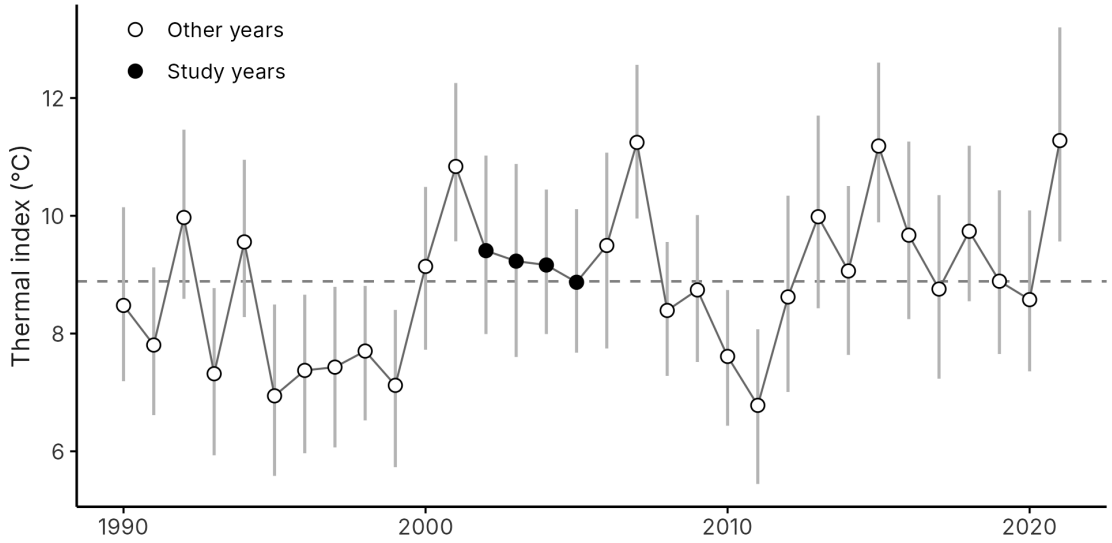


Figure A7.1: Thermal index (mean modeled temperature, 18 May–16 Jul) averaged across all surveyed reaches, 1990–2021. Bars: 10th–90th percentile among reaches. Filled points: study years. Dashed line: 1990–2021 mean.

A7.2 Is the index a property of the reach?

Yes. The order of reaches from cold to warm was almost the same in every year: the median Spearman correlation between the reach values of any two years was 0.96 (minimum 0.83), and reach means over the study years correlated 0.997 with reach means over 1990–2021. Over the full record, the index varied about as much between years network-wide (SD 1.23 °C) as among reaches (SD 1.03 °C), and the reach-by-year part was small (SD 0.25 °C; Table A7.1). The reach mean ($\bar{T}$) estimated from the study years therefore measures each reach's long-term thermal regime. The between-reach effect in the final model (Section A6) compares places that are consistently warmer or cooler, not places that happened to be warm in 2002–2005.

A7.3 How large are the within-reach anomalies?

The within-reach temperature contrast in the final model ($T - \bar{T}$) is small: SD 0.25 °C, range -0.66 to +0.44 °C. Because the model includes year effects, this contrast is estimated from the reach-specific part of each anomaly, beyond the shift shared by the whole network that year. In the study years that reach-specific part had SD 0.18 °C, comparable to its SD over the full record (0.25 °C; Figure A7.2 B). The network-wide part

Table A7.1: Variation in the thermal index among reaches, among years (network-wide) and in the reach-by-year interaction, for the full record and for the study years. Two-way decomposition on reaches with a complete record; SD in °C.

Component	SD, 1990–2021	Share, 1990–2021	SD, 2002–2005	Share, 2002–2005
Among reaches	1.03	40%	1.05	94%
Among years (network-wide)	1.23	57%	0.19	3%
Reach-by-year	0.25	2%	0.19	3%

was much narrower than in the record: anomalies from each reach's 1990–2021 mean had SD 0.28 °C in the surveyed reach-years, against 1.26 °C over all reach-years (Figure A7.2 A).

This has three consequences for interpretation:

1. The within-reach slope (7.0 d per °C, Section A6) is estimated from anomalies of a few tenths of a degree. In the longer record, network-wide anomalies of 1–2 °C were common. Applying the slope to such years would extrapolate well beyond the data.
2. The year effects in the final model cannot be attributed to temperature. The four study years were thermally similar (network means within 0.5 °C), yet adjusted mean spawn date differed by up to 7 days among years. Something other than this thermal index (for example escapement, flow or the timing of adult arrival) must underlie those differences.
3. A test of whether spawn timing tracks large between-year temperature differences would need redd surveys that include cold years (such as the late 1990s) and warm years (such as 2007, 2015 and 2021).

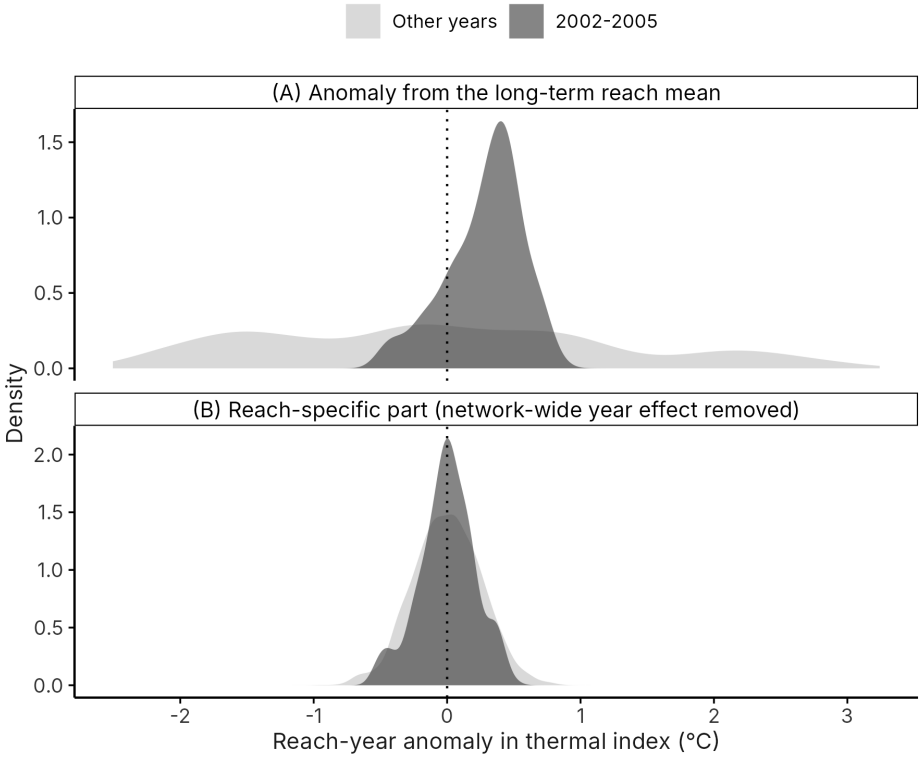


Figure A7.2: Reach-year anomalies in the thermal index, 2002–2005 (dark) vs the other years of 1990–2021 (light). (A) Anomaly from each reach's 1990–2021 mean. (B) The reach-specific part of the anomaly after removing the network-wide year effect; this is the part that informs the within-reach slope once year effects are in the model.